

Module 1: Big Ideas in Data Science Methodology

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What Data Science Really Is

Data science is not just about building models or using tools. It is a **structured way of solving problems with data** that blends:

- Business or domain understanding
- Statistical and analytical reasoning
- Data collection, preparation, and modeling
- Iteration, evaluation, and feedback

The problem and the question always come **before** the data and the model.

Why Methodology Matters

A data science methodology provides a **repeatable and defensible framework** for making decisions. Without a methodology:

- Teams jump to solutions too quickly
- Models answer the wrong question
- Results are difficult to explain or trust

Methodology helps ensure that we are solving the **right problem**, using the **right approach**, with the **right data**.

John Rollins' Data Science Methodology (10 Stages)

1. **Business Understanding**
Clearly define the problem, goals, and success criteria.
2. **Analytic Approach**
Decide how data can be used to answer the question (descriptive, predictive, classification, clustering, etc.).
3. **Data Requirements**
Identify what data is needed, in what format, and at what level of detail.
4. **Data Collection**
Gather the data from relevant sources. Missing or unavailable data is common and acceptable.
5. **Data Understanding**
Explore the data using statistics and visualization to assess quality, patterns, and gaps.
6. **Data Preparation**
Clean, transform, and engineer features to make the data usable for modeling. This stage often takes the most time.
7. **Modeling**
Build models based on the chosen analytic approach and prepared data.
8. **Evaluation**
Assess whether the model answers the original business question and meets performance expectations.
9. **Deployment**
Put the model into real-world use so stakeholders can act on the results.
10. **Feedback**
Learn from real-world performance and iterate on the model, data, or problem definition.

These stages are **iterative**, not strictly linear.

CRISP-DM Overview

CRISP-DM (Cross-Industry Standard Process for Data Mining) is a widely used industry framework with six stages:

- Business Understanding
- Data Understanding
- Data Preparation
- Modeling
- Evaluation
- Deployment

CRISP-DM closely aligns with John Rollins' methodology, with **iteration and feedback embedded throughout the process**.

Key Takeaways for This Module

- Start with **clarity**, not code
 - Data science is an **iterative process**, not a one-time workflow
 - Data preparation and understanding often consume **most of the project effort**
 - Models only create value when they are **understood, trusted, and used**
 - Feedback improves both the model **and** the original problem definition
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The Data Scientist Mindset

Think less like a tool user and more like a **problem solver**.

The goal is not to build a model — the goal is to **make better decisions with data**.